EIMS log investigation

Evidence grounded multi vendor analysis

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Cooperative education context

Enterprise infrastructure produces fragmented operational evidence.

The prototype supports traceable collection and investigation.

Graduation focus: a bounded Analyzer workflow.

Host and supervisor details require author confirmation.

The evidence problem

Logs arrive from different vendors and disconnected systems.

A repeated error message does not establish its physical cause.

Analysts need the original file, line and surrounding context.

Results must retain uncertainty when evidence is incomplete.

Current graduation objectives

Classify supported sources and preserve event identity.

Separate observed failure from job outcome and possible causes.

Retain source evidence through user-owned History.

Make evaluation and review reproducible.

System overview

FastAPI backend and PostgreSQL persistence

Next.js 16.3.0 dashboard with React 19.2.8

Secure BFF and revocable user sessions

Redis and MinIO remain part of the wider architecture.

System boundaries

Dashboard sends authenticated requests through the BFF.

Analyzer parses records and produces evidence-linked summaries.

PostgreSQL retains owner-scoped investigations.

Offline evaluation injects isolated storage and blocks external network.

Historical offline collection

September prototype used a portable Windows collector.

Collected evidence was packaged for separate ingestion.

Historical field observations are retained as provenance.

Current physical USB acceptance remains NOT_TESTED.

Historical ingestion

Evidence fingerprints supported application-level deduplication.

September observations: 49 Windows events and 10 findings.

Historical A9 display evidence is a separate runtime.

Redis and EVTX recovery work remains a separate branch stack.

Current investigation workflow

Upload synthetic Fortinet, VMware or Windows evidence.

Review source classification, symptoms and uncertainty.

Inspect the saved source excerpt with file and line coordinates.

Reopen History and check ownership in an approved runtime.

Three validation boundaries

Engineering: synthetic tests and source contracts

Scientific: authentic human judgments on frozen predictions

Runtime: the actual authenticated HTTPS browser workflow

A pass in one boundary does not establish the others.

Measured engineering results

7335aa2 baseline: 394 automated tests passed.

302 Analyzer, 46 PostgreSQL and 46 USB isolated tests

Eight synthetic counterexamples passed.

12 DEV DRAFT cases and 61 claims have zero human reviews.

Diagnostic accuracy remains NOT_MEASURED.

Five minute offline demonstration

Fortinet configuration observation: no fabricated failure

VMware CTK: technical failure, physical cause UNKNOWN

Windows 1129: connectivity symptom, missing time preserved

Open source context, then explain next evidence and History limits.

Engineering lessons

Evidence coordinates must survive serialization and History.

Last observed failure does not establish final job outcome.

RC2 hides empty failure sections and preserves Windows identity.

Review hashes prevent judgments attaching to changed predictions.

Remaining acceptance gates

Human review of Fortinet, VMware and Windows predictions

Explicitly authorized trusted HTTPS runtime

Author and institutional review of staged report and slides

Owner-approved integration of core PRs with fresh CI

Conclusion and discussion

The prototype supports traceable, bounded log investigation.

It preserves uncertainty instead of inventing a physical cause.

Current evidence supports engineering claims, not diagnostic accuracy.

Next step: genuine review and authorized runtime acceptance.